

Spectra Analysis – Spectra And Feature Extraction

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Spectra

- While some broad phonetic features can be interpreted from a waveform,
- more detailed classification requires spectral features.
- Spectral features (which are based on the insight of Fourier) are a representation of the input in terms of frequencies.
- Fourier insight: any complex wave is the sum of simple waves.

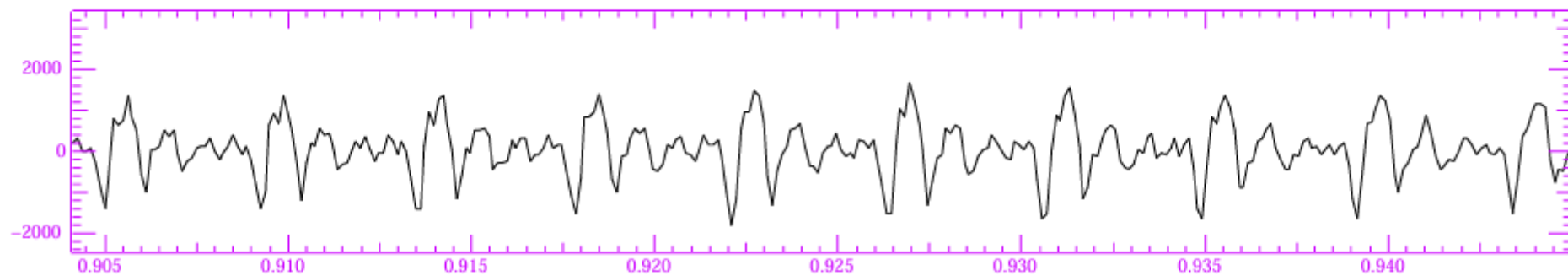


Figure 7.21 The waveform of part of the vowel [æ] from the word *had* cut out from the waveform shown in Figure 7.20.

- The picture shows the sound wave of the vowel “æ” (as in *had*) taken from a specific moment (0.9 seconds) in a spoken sentence.
- If you look at the whole image, you can see one big repeating pattern. This large pattern appears **around nine times** in the displayed time window.
- Inside each big pattern, there is a smaller pattern. You can see **four smaller peaks** inside each large repeating shape.
- Frequency = number of cycles per second. The big pattern repeats ~9 times over ~0.036 seconds. So, $9 \div 0.036 \approx$ **250 cycles per second → 250 Hz**.

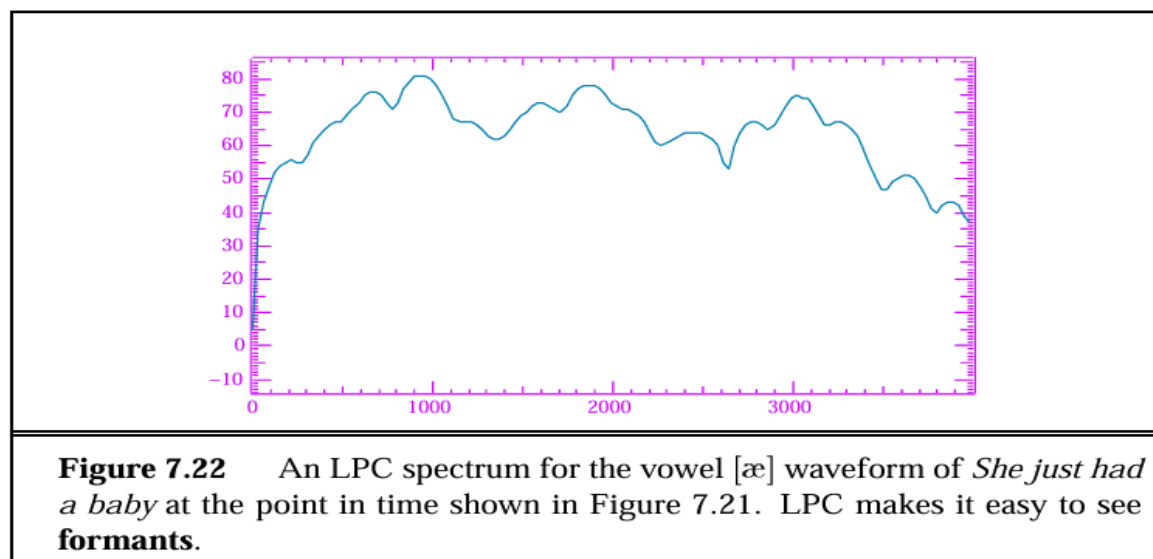
- Since there are **4 smaller waves inside each big wave**, its frequency must be **$4 \times 250 \text{ Hz} = 1000 \text{ Hz}$** .
- On the top of each medium-sized wave, you can see **even smaller ripples**—two tiny wiggles.
- Since there are **2 tiny waves** on each 1000 Hz wave, their frequency is about **$2 \times 1000 = 2000 \text{ Hz}$** .

Spectrum

- A **spectrum** is a picture that shows **all the frequencies** that make up a sound.
- A **Fourier transform** is a mathematical method that breaks a complex sound into its individual frequencies.
- In speech processing, we usually don't use the raw Fourier spectrum.
- Instead, we use a smoother version called the **LPC spectrum**.

LPC Spectrum

- LPC (**Linear Predictive Coding**) is a way of coding the spectrum, which makes it easier to see where the spectral peaks are.
- LPC helps highlight the important **peaks** in the spectrum so we can see them clearly.



- Figure 7.22 shows an LPC spectrum for the waveform in Figure 7.21.
- On the spectrum graph:
 - The **horizontal line** (x-axis) shows frequency (in Hz).
 - The **vertical line** (y-axis) shows how strong each frequency is (in dB = loudness).
- From the graph, we see big peaks around:
 - **930 Hz**
 - **1860 Hz**
 - **3020 Hz**
- These are the most important frequencies for this vowel.
- Other frequencies exist, but they're weaker.

- **Why is a spectrum useful?**
- The peaks in a spectrum tell us which speech sound it is. Every sound has its own **unique pattern of peaks**.
- For example, Analogy: Stars' light has spectral patterns that identify which elements are inside them. We can identify an element by its spectral lines.
- Similarly, speech sounds can be identified by their **spectral signature**.
- Both humans and computers rely on **spectral patterns** to recognise speech.
- Your inner ear (cochlea) naturally performs something like a **spectrum analysis** to detect frequencies.
- Computer speech recognition systems (HMM-based) also use **spectral features**, often versions of LPC spectra.

Spectrogram

- A **spectrum** shows what frequencies are present **at one moment**.
- A **spectrogram** shows **how the frequencies change over time**.
- In a spectrogram graph:
 - Horizontal axis (**x-axis**) = time
 - Vertical axis (**y-axis**) = frequency (in Hz)
- Darkness shows **how strong a frequency is**.
 - Dark = loud/strong
 - Light = weak

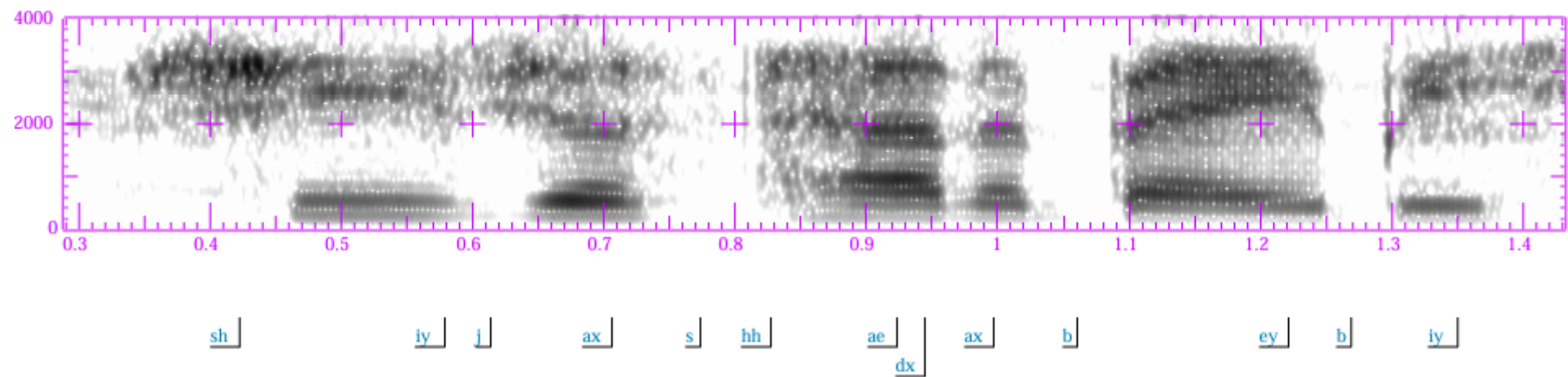
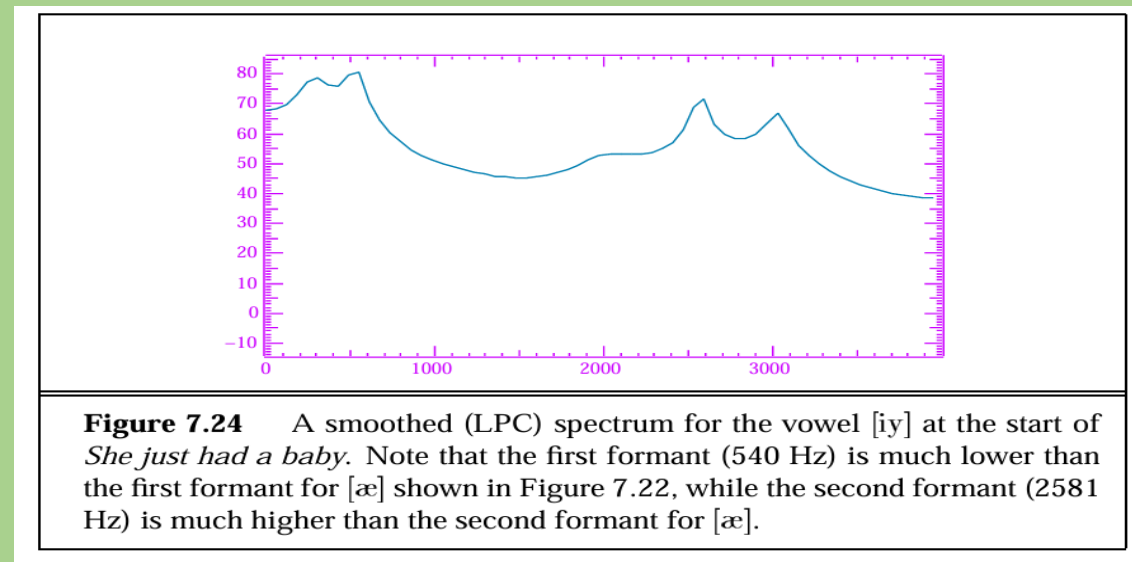


Figure 7.23 A spectrogram of the sentence “She just had a baby” whose waveform was shown in Figure 7.20. One way to think of a spectrogram is as a collection of spectra (time-slices) like Figure 7.22 placed end to end.

- In the spectrogram, near **0.9 seconds**, there is a **dark horizontal line** at **1000 Hz**.
- This means the sound has a strong frequency at 1000 Hz at that time.
- The vowel **[iy]** (as in *she*) contains a strong frequency near **1000 Hz**.
- The dark horizontal bars on a spectrogram, representing spectral peaks, usually of vowels, are called **formants**.
- Spectra help us figure out **which speech sound (phone)** is being spoken.
- Each vowel has **formants (F1, F2, F3)** at different frequencies. These formant positions help identify vowels.

- For the vowel [æ] (as in *had*), the strong resonant peaks were at :
 - F1 $\approx$ 930 Hz
 - F2 $\approx$ 1860 Hz
 - F3 $\approx$ 3020 Hz
- These values identify “[æ].”
- Now look at another vowel: [iy] (the vowel in *she*).
- Figure 7.24 displays the LPC spectrum of this vowel.



- For **[iy]**, F1 $\approx$ **540 Hz**, which is much lower than [æ]'s F1 (930 Hz).
- For **[iy]**, F2 $\approx$ **2581 Hz**, which is much higher than [æ]'s F2 (1860 Hz).
- In the spectrogram (Figure 7.23), around **0.5 seconds**, you can see **dark horizontal bands** at $\sim$ 540 Hz and $\sim$ 2581 Hz. These dark bands **are the formants** of [iy].
- F1 and F2 are the **most important** frequencies for telling vowels apart.
- But their exact values change slightly for different speakers.
- Not only vowels — other sounds like **nasals** (n, m, ŋ) and **liquid sounds** (l, r) also have distinct formant patterns.

- **Why do different vowels have different spectral signatures?**
- The mouth and throat create **resonances** (like echoes) at certain frequencies. These resonances produce **formants**.
- The mouth acts like a **filter**: Some frequencies pass strongly, and others are weakened.
- When you move your tongue, you change the **shape** and **size** of the mouth cavities
- This changes which frequencies are amplified
- This is why each vowel has different formants.

Feature Extraction

- We've looked quickly at waveforms and spectra, but now you should understand why spectral features are important and how they connect to the sound wave.
- Let's now summarise the process of extraction of spectral features, beginning with the sound wave itself and ending with a feature vector.
- **sound wave → spectral features → feature vector**
- The first step is converting the continuous sound into digital form.
- This process of **analog-to-digital conversion** has two steps: **sampling and quantisation.**

- **Sampling** measures amplitude at a particular time
- **Sampling rate** is the number of samples taken per second.
- Common sampling rates are 8,000 Hz and 16,000 Hz.
- To accurately measure a wave, it is necessary to have at least two samples in each cycle
 - one measuring the positive part of the wave and
 - one measuring the negative part.
- Taking more samples gives a more accurate picture of the wave.
- But less than two samples will cause the frequency of the wave to be completely missed.
- You need **2 samples for each wave cycle**, so the fastest wave you can measure has a frequency equal to **half of the sampling rate**.

- The maximum frequency for a given sampling rate is called the **Nyquist frequency**
- **Nyquist frequency = half of the sampling rate.**
- It is the **highest frequency** that can be correctly measured when you digitise a sound.
- **Nyquist frequency = sampling rate / 2**
- Most information in human speech is in frequencies below 10,000 Hz
- Thus, a 20,000 Hz sampling rate would be necessary for complete accuracy.
- **Sampling rate = 2 x maximum frequency**
- Sampling rate = $2 \times 10,000 = 20,000$ Hz

- Telephone speech is filtered by the switching network, and only frequencies less than 4,000 Hz are transmitted by telephones.
- Thus, an 8,000 Hz sampling rate is sufficient for telephone-bandwidth speech like the Switchboard corpus.
- Even an 8,000 Hz sampling rate requires 8000 amplitude measurements for each second of speech, and so it is important to store the amplitude measurements efficiently.
- Samples are usually stored as integers, either 8-bit (values from –128 to 127) or 16-bit (values from –32768 to 32767).
- This process of representing a real-valued number as an integer is called **quantization**.

- Once a waveform has been digitized, it is converted to some set of spectral features
- An LPC spectrum is represented by a vector of features; each formant is represented by two features, plus two additional features to represent spectral tilt.
- Thus, 5 formants can be represented by 12 ($5 \times 2 + 2$) features.
- It is possible to use LPC feature and **feed it directly** into the HMM as the input for speech recognition.
- Even though LPC features can be used directly, **we usually don't stop there.**

- Because
 - We can make the features **more stable**
 - More **compact**
 - More **noise-resistant**
 - More **similar to how humans hear**
- Therefore, extra steps (transformations) are often applied to make the LPC features better for speech recognition.
- One popular feature set is **Cepstral coefficients**, which are computed from the LPC coefficients by taking the Fourier transform of the spectrum.
- Cepstral coefficients are used to extract the important features of the speech signal so that the computer can recognize the sound more easily.

- Another feature set, **PLP (Perceptual Linear Predictive analysis)**, modifies them in ways consistent with human hearing.
- PLP keeps the parts of the sound that humans hear clearly and removes the parts that humans hear poorly.
- Our ears hear high frequencies poorly and hear loudness in a compressed way.
- So PLP applies various filters to the LPC spectrum and takes the cube root of the features.

THANKYOU

for you patience